Miles Waugh

Irvine, CA | contact@mileswaugh.com | (408) 470-0583 | miles-waugh | piano-miles

Education

BS **University of California, Irvine**, double major: Computational Physics and Computational & Applied Mathematics, fourth-year undergrad — Sep 2023 to present
GPA: 3.93

- **Selected Coursework:** Graduate Quantum Mechanics, Graduate Condensed Matter Theory, Real Analysis, Complex Analysis, Numerical Analysis

Research Experience

Caltech SURF Program, Summer Undergraduate Researcher, Anima Lab — Pasadena, CA
Summer 2025, 2026

Summer 2026

- Extended prior research to incorporate electron-phonon interactions, learning continuous-time dynamics via parameter-conditioned Fourier Neural Operators (FNOs) in PyTorch.
- Derived and implemented a Padé-based local propagator with mathematically enforced derivative-matching conditions.
- Models captured global trends but not local fluctuations, suggesting the reduced representation cannot determine a unique propagator.

Summer 2025

- Trained FNOs on nonequilibrium Mott insulator dynamics in PyTorch to extrapolate toward the thermodynamic limit.
- Ran large-scale GPU jobs with SLURM, conducted ablation studies, and benchmarked against vision transformers.
- Achieved $\sim$1500$\times$ faster runtimes with FNOs, scaling to 2048$\times$2048 lattices and generalizing far beyond the training regime.
- Results provide a novel pathway to probe avalanche photoexcitations relevant to solar cell efficiencies beyond the classical limit.

UCI, Whiteson Lab, Undergraduate Researcher — Irvine, CA
Nov 2023 to May 2026

- Conducting high-energy physics research with the ATLAS experiment at CERN to investigate dark matter production from W' bosons.
- Generated large-scale simulated collision data using MadGraph, Pythia8, and Delphes, modeling event generation, hadronization, and detector responses.
- Trained a transformer-based machine learning model on simulated data in Python using NVIDIA A100 GPUs on a high-performance computing (HPC) cluster, outperforming the standard χ^2-based method.

European Organization for Nuclear Research (CERN), Visiting Summer Student — Geneva, Switzerland
Summer 2024

- Continued research on reconstructing particle decays at CERN's main site as part of the ATLAS experiment.
- Attended lectures by leading scientists on experimental and theoretical particle physics.

Professional Experience

Pandora Bio, Inc, Data Science Intern — Sunnyvale, CA
May 2023 to Sep 2024

- Built interactive visualizations for a cross-platform mobile app using Flutter and Dart to monitor emotional health with collected smartphone data.
- Implemented ML models with Python to extract and process user app usage data on AWS.

theCoderSchool, Computer Science Instructor — Cupertino, CA
May 2022 to August 2023

- Designed and instructed classes and workshops for students 7-17 in Scratch, Python, HTML/CSS/JS, and Java, practicing various topics including game and software

development, computer simulations, and neural networks.

Delfi Diagnostics, Clinical Data Science Intern — Palo Alto, CA
Jun 2021 to Jul 2021

- Developed interactive R Shiny dashboards to visualize clinical trial data, including patient history and blood sample metrics, enabling consistent quality monitoring for early-stage cancer detection.

Publications & Patents

Waugh, M., et al. (in preparation). SPANet for Dark Matter with Hadronically Decaying W'. — Paper in preparation

Waugh, M., et al. (2026). Toward the Thermodynamic Limit: Neural Operators for Non-equilibrium Dynamics of Mott Insulators. *arXiv preprint* arXiv:2602.19484. — Paper, 2026
Accepted to *ICLR 2026 Workshop on Artificial Intelligence and Partial Differential Equations* (poster).

Waugh, M. (2025, November). *Toward the Thermodynamic Limit: Neural Operators for Non-equilibrium Dynamics of Mott Insulators* [Poster]. Caltech & University of Chicago Conference on AI+Science, Pasadena, CA. — Poster, 2025

Tara Maddala, Megan Rothney, Jason Carlson, Akhil Nistala, ***Miles Waugh***, Daniel Civello, Jennifer Geis (2024). *Early detection tools for mental health* (U.S. Patent Application No. 63/616,140). — Patent, 2024

Honors & Awards

Annual Resonance Fellow Award, UCI Department of Physics & Astronomy. Awarded for mentorship of incoming physics students through the Resonance program. — 2025

Annual Physics & Astronomy Peer Choice Award for Non-Graduating Majors, UCI. Selected by fellow physics majors for enthusiasm and passion for physics. — 2025

Dean's List, University of California, Irvine. — All quarters to date

CERN Research Opportunity, selected by Professor Whiteson to conduct high-energy particle physics research at CERN.

Teaching, Mentorship & Outreach

UCI, Introduction to Programming for Numerical Analysis, Learning Assistant — Irvine, CA
Spring 2026

- Assisted students with course material and practice problems in numerical analysis and scientific programming.
- Developed a supplementary Jupyter notebook problem set exploring Newton's method for numerical root-finding.

Undergraduate Mathematics Committee, UCI, Study Jams Tutor — Irvine, CA
Fall 2026

- Volunteered as a tutor for differential equations and numerical analysis, helping students work through problems and derivations during open study sessions.

UCI Resonance Program, Peer Mentor — Irvine, CA
Fall 2024, 2026

- Mentored incoming physics freshmen and transfer students on course planning, academic goals, and preparation for graduate school and professional careers.

Society of Physics Students (SPS), Outreach Volunteer — Irvine, CA
Spring 2026

- Visited a local high school to present on studying physics and undergraduate research, and led interactive physics demonstrations for students.

Academic & Professional Engagement

- **Undergraduate Mathematics Committee, UCI** Member (2025–present)
- **Quantum Computing @ UCI** Project Lead (2023)
  Led a small team developing a variational quantum eigensolver (VQE) to approximate ground states of simple molecules using Qiskit; the project was exploratory in scope.
- **Society of Physics Students (SPS)** Member (2023–present)
- **Astronomy Club, UCI** Member (2023–present)
- **TKA FTC Robotics Team** Member and Hardware Lead (2020–2022)
  Designed and maintained robot hardware; awarded the *Hardware Excellence Award (2021)* for mechanical design and innovation.
- **TKA Math Club & Coding Club** President (2021–2022)

Technical Skills & Languages

Programming:

- **Advanced:** Python, Mathematica, Processing, P5js
- **Intermediate:** C++, MATLAB, Java, ROOT, R, HTML/CSS/JavaScript

Frameworks/Tools: PyTorch, NumPy, SciPy, Matplotlib, TensorFlow, LaTeX, Git, Linux, SLURM, AWS
Methods: Numerical methods, optimization, machine learning, Monte Carlo simulation, data analysis, HPC workflows
Interests: tennis, cooking, piano